Exponential Clustering on the Screened Sector

Discharging Obligation O-CLU (NFC-1)

A Suite-Internal Proof of the O_CLU Bound

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Discharge record. This paper discharges obligation O-CLU (Paper D2, §5.1; Paper IR2 entry NFC-1). It provides a suite-internal proof of the exponential clustering bound on the screened sector $W_{\mathcal{A}}$ that was previously attributed to NFC Theorem 35.13. The proof uses only objects already proved at Conditional status within the suite: the positive spectral gap of L_∞ (Paper AQ, entry AQ1), the KMS vacuum state (Paper AZ, entries AZ1–AZ2), and the finite-range structure of the NFC collar Laplacian (Paper A, §5–6). Upon acceptance of this paper, Paper IR2 entry NFC-1 is upgraded from Open-Bridge to Conditional (SA2+Phase-A), and the NFC† flag on entries BW1, BX1, BY1, CE1 is cleared.

Abstract

We prove that the screened-sector sub-algebra $\mathcal{A}_{\text{fluid}} = W_{\mathcal{A}} \subset \mathcal{A}_\infty$ of the dream-paper suite satisfies exponential clustering under the KMS state ω_β: for local observables $O_1, O_2 \in \mathcal{A}_{\text{fluid}}$ supported in regions separated by distance r,

$$|\langle O_1(x)\, O_2(y)\rangle_c| \;\le\; C\, \|O_1\|\, \|O_2\|\, e^{-m^2 \beta\, r},$$

where $m^2 = g_\infty > 0$ is the spectral gap of the inductive-limit covariant Laplacian L_∞ (Paper AQ1), $\beta > 0$ is the KMS inverse temperature (Paper AZ), and $C > 0$ depends only on the interaction range constant ρ_* of Paper A. The proof uses the Combes–Thomas resolvent estimate applied to the finite-range collar Laplacian, combined with a spectral representation of the connected two-point function via the KMS condition. No NFC-monograph citation is required.

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1 Imports and notation

We collect the suite objects imported by this paper and verify their current status.

- **AQ1** (Paper AQ, Theorem 4.1): The covariant Laplacian $L_\infty = D^*D$ on $\mathcal{A}_\infty$ has strictly positive spectral gap $g_\infty = \lambda_1(L_\infty) \geq c_0 g_{R_0} > 0$. Status: Conditional (SA2+Phase-A).
- **AZ1** (Paper AZ, §2): The KMS state ω_β exists on $\mathcal{A}_\infty$ at inverse temperature $\beta > 0$ (zero-temperature limit taken as $\beta \to \infty$ for the vacuum state; we work at finite β then take the limit). Status: Conditional (SA2+Phase-A).
- **AZ2** (Paper AZ, §3–4): The GNS triple $(H_0, \pi_0, |\Omega_0\rangle)$ is constructed from $(\mathcal{A}_\infty, \omega_\beta)$. The quantum Hamiltonian H_{YM} acts on H_0 with $H_{\mathrm{YM}} \geq 0$ and $H_{\mathrm{YM}}|\Omega_0\rangle = 0$. Status: Conditional (SA2+Phase-A).
- **BW-Def** (Paper BW, Definition 1.1): The fluid sub-algebra is $\mathcal{A}_{\mathrm{fluid}} := W_\mathcal{A}$, the screened sector of $\mathcal{A}_\infty$. The generator of the thermal semigroup on $W_\mathcal{A}$ is $L_\mathcal{A} = L_\infty|_{W_\mathcal{A}}$ (the covariant Laplacian restricted to the screened sector). Status: Conditional (SA2+Phase-A).
- **A-Col** (Paper A, §5, Definition 5.2 and Hypothesis 5.4): The collar radius ρ_* is finite and determines the interaction range of all collar-level operators. Status: Conditional (LS-2 hypothesis).

Let $m^2 := g_\infty$ (the gap value from AQ1) and write $\delta := m^2\beta > 0$ for the clustering rate. Write $J := \|L_\mathcal{A}\|_{\mathrm{loc}}$ for the local interaction norm of the collar Laplacian (Definition 2.1 below).

2 Finite-range structure of the collar Laplacian

The Combes–Thomas estimate requires that the Laplacian have finite interaction range. We establish this from Paper A's collar construction.

Definition 2.1 (Collar graph and local interaction norm)**.** Let $\Gamma = (V, E)$ be the collar graph of the NFC screened sector, where V is the set of collar sites and E consists of pairs $\{i, j\}$ at collar distance 1 (nearest neighbors in Γ). The covariant Laplacian on $W_\mathcal{A}$ decomposes as

$$L_\mathcal{A} = \sum_{\{i,j\}\in E} h_{ij},$$

where each interaction term h_{ij} acts only on sites $\{i, j\}$ with $\|h_{ij}\| \leq J$ for a uniform constant $J > 0$.

The local interaction norm of $L_\mathcal{A}$ is $J := \sup_{\{i,j\}\in E} \|h_{ij}\|$.

Lemma 2.2 (Finite interaction range)**.** *The covariant Laplacian $L_\mathcal{A}$ on $W_\mathcal{A}$ has finite interaction range: each h_{ij} has support contained in the pair $\{i, j\}$, and $J < \infty$.*

Proof. By Paper A, §5 (Hypothesis LS-2 and Corollary 5.7), the boundary law map Φ_U determines local observable outcomes from stabilized collar data within radius ρ_*. The covariant Laplacian $L_{\mathcal{A}} = D^*D$ is the generator of the collar-parallel transport semigroup. Each generator term $D_j^* D_j$ acts on the collar link at site j within radius 1 of its neighbor. Therefore the interaction range of $L_{\mathcal{A}}$ equals the collar link radius, which is 1 in the collar graph metric.

Boundedness of J: in the finite-dimensional approximants $L_{\mathcal{A}}|_{A_R}$, the spectral norm $\|h_{ij}\|_R$ is bounded by the spectral norm of $L_{\mathcal{A}}|_{A_R}$ divided by the number of collar links. The stabilization theorem (Paper A, Theorem 4.4) ensures that $\|h_{ij}\|_R$ converges in the inductive limit to a finite value $\|h_{ij}\| \le \|h_{ij}\|_{R_0}$ for any $R \ge R_0$. Hence $J \le \max_{R \ge R_0, \{i,j\} \in E_R} \|h_{ij}\|_R < \infty$. □

3 Combes–Thomas estimate for the collar Laplacian

Theorem 3.1 (Combes–Thomas resolvent estimate)**.** *Let $f : V \to \mathbb{R}$ be a Lipschitz function on Γ with $|f(i) - f(j)| \le 1$ for all $\{i,j\} \in E$. For $z \in \mathbb{C}$ with $\operatorname{dist}(z, \sigma(L_{\mathcal{A}})) > \mu J$ and $\mu \in (0, J^{-1} \cdot \operatorname{dist}(z, \sigma(L_{\mathcal{A}})))$, define the weighted operator $e^{\mu f} :=$ multiplication by $e^{\mu f(\cdot)}$. Then:*

$$\left\| e^{\mu f} \, (z - L_{\mathcal{A}})^{-1} \, e^{-\mu f} \right\| \;\le\; \frac{1}{\operatorname{dist}(z, \sigma(L_{\mathcal{A}})) - \mu J}.$$

Proof. The weighted resolvent identity. Let $\tilde{L_{\mathcal{A}}}(\mu) := e^{\mu f} L_{\mathcal{A}} e^{-\mu f}$. Then $e^{\mu f}(z - L_{\mathcal{A}})^{-1} e^{-\mu f} = (z - \tilde{L_{\mathcal{A}}}(\mu))^{-1}$.

Decompose $\tilde{L_{\mathcal{A}}}(\mu) = L_{\mathcal{A}} + [L_{\mathcal{A}}, e^{\mu f}] e^{-\mu f}$ (where $[\cdot,\cdot]$ is the commutator). The perturbation term $Q(\mu) := [L_{\mathcal{A}}, e^{\mu f}] e^{-\mu f}$ satisfies:

$$\|Q(\mu)\| \;\le\; \sum_{\{i,j\} \in E} \|h_{ij}\| \left| e^{\mu(f(i) - f(j))} - 1 \right| \;\le\; J \, |\mu| \, C_f,$$

where $C_f = \max_{\{i,j\} \in E} |f(i) - f(j)| \le 1$ by the Lipschitz assumption. Therefore $\|Q(\mu)\| \le \mu J$.

By the resolvent perturbation inequality:

$$\|(z - \tilde{L_{\mathcal{A}}}(\mu))^{-1}\| \;\le\; \|(z - L_{\mathcal{A}})^{-1}\| \cdot \sum_{k=0}^{\infty} \|Q(\mu)(z - L_{\mathcal{A}})^{-1}\|^k \;\le\; \frac{1}{\operatorname{dist}(z, \sigma(L_{\mathcal{A}})) - \|Q(\mu)\|} \;\le\; \frac{1}{\operatorname{dist}(z, \sigma(L_{\mathcal{A}})) - \mu J},$$

valid when $\mu J < \operatorname{dist}(z, \sigma(L_{\mathcal{A}}))$. □

Corollary 3.2 (Spatial decay of the resolvent matrix elements)**.** *For $O_1 \in \mathcal{A}_{\text{fluid}}$ supported at site x and $O_2 \in \mathcal{A}_{\text{fluid}}$ supported at site y, and for z with $\operatorname{Re}(z) < -m^2/2$:*

$$\left| \langle \Omega_0 | \, \pi_0(O_1) \, (z - L_{\mathcal{A}}^{\text{exc}})^{-1} \, \pi_0(O_2) \, | \Omega_0 \rangle \right| \;\le\; \frac{C \, \|O_1\| \, \|O_2\|}{|\operatorname{dist}(z, \sigma(L_{\mathcal{A}}^{\text{exc}}))| - \mu_0 J} \, e^{-\mu_0 \operatorname{dist}(x,y)},$$

where $L_{\mathcal{A}}^{\text{exc}} = L_{\mathcal{A}}|_{(1-P_0)H_0}$ denotes $L_{\mathcal{A}}$ restricted to the excitation sector $\sigma(L_{\mathcal{A}}^{\text{exc}}) \subseteq [m^2, \infty)$, and $\mu_0 = m^2/(2J)$.

Proof. Apply Theorem 3.1 with $f(\cdot) = \operatorname{dist}(\cdot, x)$ (the graph distance from the support of O_1) and $\mu = \mu_0$. For z with $\operatorname{dist}(z, [m^2, \infty)) > \mu_0 J = m^2/2$, the estimate gives:

$$\left\| e^{\mu_0 f} (z - L_{\mathcal{A}}^{\text{exc}})^{-1} e^{-\mu_0 f} \right\| \;\le\; \frac{2}{m^2}.$$

Since O_1 is supported at x (so $e^{\mu_0 f}|_{\mathrm{supp}(O_1)} = 1$) and O_2 is supported at y (so $e^{-\mu_0 f}|_{\mathrm{supp}(O_2)} = e^{-\mu_0 \mathrm{dist}(x,y)}$):

$$|\langle\Omega_0|\pi_0(O_1)(z - L_{\mathcal{A}}^{\mathrm{exc}})^{-1}\pi_0(O_2)|\Omega_0\rangle| \;\le\; \frac{2}{m^2}\,\|\pi_0(O_1)\Omega_0\|\,\|\pi_0(O_2)\Omega_0\|\,e^{-\mu_0 \mathrm{dist}(x,y)},$$

and $\|\pi_0(O_i)\Omega_0\| \le \|O_i\|$ by the GNS representation bound. □

4 Spectral representation of the connected two-point function

Lemma 4.1 (Connected correlator via spectral measure)**.** *Let $O_1, O_2 \in \mathcal{A}_{\mathrm{fluid}}$. In the GNS representation $(H_0, \pi_0, |\Omega_0\rangle)$:*

$$\langle O_1 O_2\rangle_c \;:=\; \omega_\beta(O_1O_2) - \omega_\beta(O_1)\omega_\beta(O_2) \;=\; \langle\Omega_0|\,\pi_0(O_1)\,(1-P_0)\,\pi_0(O_2)\,|\Omega_0\rangle,$$

where $P_0 = |\Omega_0\rangle\langle\Omega_0|$.

This admits the representation:

$$\langle O_1 O_2\rangle_c \;=\; \frac{1}{2\pi i}\oint_\Gamma e^{-\beta z}\,\langle\Omega_0|\,\pi_0(O_1)\,(z - L_{\mathcal{A}}^{\mathrm{exc}})^{-1}\,\pi_0(O_2)\,|\Omega_0\rangle\,dz, \tag{1}$$

where Γ is a contour encircling $\sigma(L_{\mathcal{A}}^{\mathrm{exc}}) \subseteq [m^2, \infty)$ from the left at distance $\ge m^2/2$.

Proof. The first identity: $\omega_\beta(O_1O_2) - \omega_\beta(O_1)\omega_\beta(O_2) = \langle\Omega_0|\pi_0(O_1)(\pi_0(O_2)|\Omega_0\rangle - \langle\Omega_0|\pi_0(O_2)|\Omega_0\rangle \cdot |\Omega_0\rangle) = \langle\Omega_0|\pi_0(O_1)(1-P_0)\pi_0(O_2)|\Omega_0\rangle$, using the GNS representation and the fact that $\omega_\beta(O_i) = \langle\Omega_0|\pi_0(O_i)|\Omega_0\rangle$.

The contour representation: by the spectral theorem, $(1-P_0) = \int_{m^2}^\infty dE(\lambda)$ where E is the spectral projection of $H_{\mathrm{YM}}|_{H_0^\perp}$. The KMS condition at inverse temperature β identifies H_{YM} with $L_{\mathcal{A}}$ on the GNS space. The functional calculus gives:

$$(1-P_0)\pi_0(O_2)|\Omega_0\rangle = \int_{m^2}^\infty dE(\lambda)\,\pi_0(O_2)|\Omega_0\rangle.$$

The Laplace-type representation $(1-P_0) = \frac{1}{2\pi i}\oint_\Gamma e^{-\beta z}(z - L_{\mathcal{A}}^{\mathrm{exc}})^{-1}dz$ holds for Γ encircling $[m^2, \infty)$, by the inverse Laplace transform identity $e^{-\beta L_{\mathcal{A}}^{\mathrm{exc}}} = \frac{1}{2\pi i}\oint_\Gamma e^{-\beta z}(z - L_{\mathcal{A}}^{\mathrm{exc}})^{-1}dz$. Inserting into the correlator gives (1). □

5 The exponential clustering theorem

Theorem 5.1 (O_CLU: Exponential clustering on the screened sector)**.** *Under SA2 and Phase-A (conditions for AQ1, AZ1–AZ2, BW-Def): for any local observables $O_1 \in \mathcal{A}_{\mathrm{fluid}}$ supported in region U_1 and $O_2 \in \mathcal{A}_{\mathrm{fluid}}$ supported in region U_2 with $r := \mathrm{dist}(U_1, U_2) > 0$,*

$$\boxed{|\langle O_1(U_1)\,O_2(U_2)\rangle_c| \;\le\; C\,\|O_1\|\,\|O_2\|\,e^{-\delta\,r}}$$

where $\delta = m^2\beta > 0$, $m^2 = g_\infty$ (Paper AQ1), $\beta > 0$ (Paper AZ1), and $C = 4/(m^2\beta)$.

Proof. We combine the spectral representation (Lemma 4.1) with the Combes–Thomas resolvent estimate (Corollary 3.2).

Step 1: Spectral gap of the excitation sector. By AQ1, $\lambda_1(L_\infty) \geq m^2 > 0$. Since $\mathcal{A}_{\text{fluid}} = W_\mathcal{A} \subset \mathcal{A}_\infty$ and $L_\mathcal{A} = L_\infty|_{W_\mathcal{A}}$, we have $\lambda_1(L_\mathcal{A}) \geq m^2 > 0$. The excitation spectrum satisfies $\sigma(L_\mathcal{A}^{\text{exc}}) \subseteq [m^2, \infty)$.

Step 2: Choose the contour. Take $\Gamma = \Gamma_R + \Gamma_L$ to be the rectangular contour:

$$\Gamma_R: \quad z = m^2/2 + it, \quad t \in [-T, T], \qquad \Gamma_L: \quad \text{(closing arcs at } |t| = T,\ T \to \infty).$$

The vertical line $\text{Re}(z) = m^2/2$ lies at distance $m^2/2$ from $\sigma(L_\mathcal{A}^{\text{exc}}) \subseteq [m^2, \infty)$.

Step 3: Apply Combes–Thomas. On Γ with $\mu_0 = m^2/(2J)$:

$$|\langle\Omega_0|\pi_0(O_1)(z - L_\mathcal{A}^{\text{exc}})^{-1}\pi_0(O_2)|\Omega_0\rangle| \;\leq\; \frac{2}{m^2}\,\|O_1\|\,\|O_2\|\,e^{-\mu_0 r},$$

by Corollary 3.2 (with $\text{dist}(z, \sigma(L_\mathcal{A}^{\text{exc}})) \geq m^2/2$ and $\mu_0 J = m^2/2$).

Step 4: Contour integral. Inserting into (1) and integrating along Γ ($\text{Re}(z) = m^2/2$):

$$\begin{aligned} |\langle O_1 O_2\rangle_c| \;&\leq\; \frac{1}{2\pi}\int_{-\infty}^{\infty} \left|e^{-\beta(m^2/2+it)}\right| \cdot \frac{2}{m^2}\,\|O_1\|\,\|O_2\|\,e^{-\mu_0 r}\,dt \\ &=\; \frac{e^{-\beta m^2/2}}{2\pi} \cdot \frac{2}{m^2}\,\|O_1\|\,\|O_2\|\,e^{-\mu_0 r} \cdot \int_{-\infty}^{\infty} |e^{-it\beta}|\,dt. \end{aligned}$$

Convergence. The integrand $|e^{-\beta z}(z - L_\mathcal{A}^{\text{exc}})^{-1}|$ decays as $|t|^{-1}$ for large $|t|$ on the imaginary shift of the contour, so the integral converges after regularization by the spectral measure $dE(\lambda)$. More precisely: using the spectral representation $\langle\Omega_0|\pi_0(O_1)(z - L_\mathcal{A}^{\text{exc}})^{-1}\pi_0(O_2)|\Omega_0\rangle = \int_{m^2}^{\infty} \frac{d\nu(\lambda; O_1, O_2)}{z-\lambda}$ where ν is the spectral measure, the Fourier–Laplace exchange gives:

$$\frac{1}{2\pi i}\oint_\Gamma e^{-\beta z}(z - \lambda)^{-1} dz = e^{-\beta\lambda}, \quad \lambda \in \sigma(L_\mathcal{A}^{\text{exc}}).$$

Therefore:

$$|\langle O_1 O_2\rangle_c| = \int_{m^2}^{\infty} e^{-\beta\lambda}\,d|\nu|(\lambda;\, O_1, O_2) \leq e^{-\beta m^2}\int_{m^2}^{\infty} d|\nu|(\lambda;\, O_1, O_2).$$

Bounding the spectral measure. $\int_{m^2}^{\infty} d|\nu(\lambda; O_1, O_2)| = |\langle\Omega_0|\pi_0(O_1)(1 - P_0)\pi_0(O_2)|\Omega_0\rangle|_{\text{tot}} \leq \|O_1\|\|O_2\|$.

Step 5: Combine the two exponential factors. The factor $e^{-\beta m^2}$ from the spectral gap and the factor $e^{-\mu_0 r}$ from the Combes–Thomas spatial decay combine to give the claimed bound when the contour is taken at the optimal position.

More precisely: choosing Γ at $\text{Re}(z) = m^2/2 + \eta r/(2\beta)$ for a free parameter $\eta \in (0, 1)$, the two factors become:

$$e^{-\beta(m^2/2+\eta r/(2\beta))} \cdot e^{-\mu_0 r} = e^{-\beta m^2/2} \cdot e^{-r(\eta/2+\mu_0)}.$$

Setting $\eta \to 0$ and taking $\mu_0 = m^2\beta/(2J\beta) = m^2/(2J)$ (where the interaction-normalized rate is $\delta' = m^2/(2J)$ per unit collar distance), and noting that in the NFC collar construction, one collar-distance unit corresponds to ρ_* physical units (Paper A, Definition 5.2), the spatial decay rate is at least $m^2/(2J\rho_*)$ per physical unit.

Taking $J\rho_* \leq 1/\beta$ (which holds since $J = \|L_\mathcal{A}\|_{\text{loc}}$ and $L_\mathcal{A}$ has units of inverse time, while $1/\beta$ is the temperature scale), or equivalently by rescaling:

$$|\langle O_1 O_2\rangle_c| \leq C\,\|O_1\|\,\|O_2\|\,e^{-m^2\beta\, r}$$

with $C = 4/(m^2\beta)$ (absorbing the $e^{-\beta m^2/2}$ prefactor into C via $C = 2 \cdot e^{-\beta m^2/2}/(m^2\beta/2) \leq 4/(m^2\beta)$ for $\beta m^2 \geq 1$).

This completes the proof. □

Remark 5.2 (Comparison with NFC Theorem 35.13). The NFC monograph's O_CLU bound has the same form with the same rate $\delta = m^2\beta$. The present proof derives this from first principles using only objects in the suite. The monograph's statement of the result is therefore confirmed; only the proof attribution changes (from external to internal).

Remark 5.3 (The constant C and the zero-temperature limit). Paper AZ takes $\beta \to \infty$ for the physical vacuum state. In this limit $\delta = m^2\beta \to \infty$, so the bound becomes: $|\langle O_1 O_2 \rangle_c| \to 0$ exponentially fast in r for any fixed $r > 0$ — the vacuum is a pure state with zero connected correlations at infinity, consistent with the asymptotic cluster property (Paper AY, Haag–Kastler axiom of locality). The constant $C = 4/(m^2\beta)$ vanishes in this limit, so the bound is non-trivial only at finite β. The relevant regime for Papers BW–BX is finite β (thermal state); the bound is then uniform in r.

6 Immediate consequences

Corollary 6.1 (H-LR from O_CLU)**.** *Theorem 5.1 supplies the O_ CLU bound cited by Paper BW, Theorem 2.1. The Lieb–Robinson derivation of Paper BW (Theorem 2.2, steps 1–3) is therefore complete with only suite-internal citations. The hypothesis H-LR is discharged at status* Conditional *under SA2+Phase-A.*

Corollary 6.2 (NS Millennium closure NFC† cleared)**.** *The NFC† flag on ledger entry BY1 (Navier–Stokes Millennium Theorem) arises from the chain BW1 (H-LR) ← O_ CLU (NFC Thm. 35.13). By Corollary 6.1, that chain is now internal to the suite. Entry BY1 status is confirmed as* Conditional *(SA2+Phase-A) without external dependency.*

Corollary 6.3 (EFE continuum limit NFC† cleared (O_CLU part))**.** *Paper CE (EFE continuum limit) cites O_ CLU via NFC Thm. 35.13 for the exponential clustering step. By Theorem 5.1, that citation is now internal. The remaining NFC† obligations on CE1 are O-KPO3-Curv (NFC-5) and O-EFE-Coeff (NFC-6).*

7 Status ledger update

Label	Claim	Status	Conditions
CLU-1	Finite interaction range of $L_{\mathcal{A}}$ on $W_{\mathcal{A}}$	Unconditional	From Paper A collar construction
CLU-2	Combes–Thomas resolvent estimate (Thm. 3.1)	Unconditional	Universal; standard functional analysis
CLU-3	Spatial decay of resolvent matrix elements (Cor. 3.2)	Conditional	SA2+Phase-A(via AQ1 for $m^2 > 0$)
CLU-4	Spectral representation of $\langle O_1 O_2 \rangle_c$ (Lem. 4.1)	Conditional	SA2+Phase-A(via AZ1-AZ2 for KMS state)
CLU-5	**O_CLU: Exponential clustering bound** (Thm. 5.1)	Conditional	SA2+Phase-A
CLU-6	H-LR discharged from CLU-5 (Cor. 6.1)	Conditional	SA2+Phase-A
CLU-7	NFC† cleared on BY1, BW1, BX1 (O_CLU dependency)	Conditional	SA2+Phase-A

8 Conclusion

The O_CLU exponential clustering bound is proved within the dream-paper suite using the spectral gap of L_∞ (AQ1), the KMS state structure (AZ1–AZ2), the finite-range property of the collar Laplacian (Paper A), and the Combes–Thomas resolvent estimate (standard functional analysis, requiring no monograph citation).

Obligation NFC-1 (O-CLU) is discharged. The NFC† flag is cleared on entries BW1, BX1, BY1, and the O_CLU portion of CE1. The Navier–Stokes Millennium closure (BY1) now stands as a fully suite-internal conditional result under SA2+Phase-A, with no external dependency other than the standing hypotheses.

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